HTML Full Notes

Simple explanations, one topic at a time: from first tags to forms, media, semantic HTML and accessibility

Getting Started

What is HTML?

HTML means HyperText Markup Language. It gives a web page its structure: headings, text, images, links and more. It is not a programming language.

```
<h1>Hello World</h1>
```

Basic page structure

Every page has the same skeleton. head holds settings. body holds what people see.

```
<!DOCTYPE html>
<html lang="en">
<head><title>My Page</title></head>
<body>Hi!</body>
</html>
```

Tags & elements

A tag is a word inside angle brackets. Most come in pairs: opening and closing. Together with the content they make an element.

```
<p>Text</p>
<!-- p = tag -->
```

Attributes & comments

Attributes add extra info to a tag, written as name="value". Comments are notes the browser does not show.

```
<a href="a.html">Link</a>
<!-- my note -->
```

Text Formatting

Headings

h1 is the biggest and most important. h6 is the smallest. Use one h1 per page.

```
<h1>Title</h1>
<h2>Part</h2>
```

Paragraphs & breaks

p makes a paragraph. br breaks a line. hr draws a line across the page.

```
<p>Hello</p>
Line<br>Next line
<hr>
```

Bold, italic & more

strong and em mean important and emphasized. mark highlights. sub and sup shrink text up or down. del crosses out.

```
<strong>Bold</strong> <em>Italic</em>
<mark>Hot</mark> H<sub>2</sub>O
```

Quotes & code

blockquote is a long quote. q is a short one. code shows code. pre keeps spaces and new lines.

```
<blockquote>Quote</blockquote>
<pre><code>x = 1</code></pre>
```

Links

Basic link

The a tag makes a link. href is where it goes.

```
<a href="https://example.com">Visit</a>
```

Open in new tab

target="_blank" opens a new tab. Add rel="noopener" for safety.

```
<a href="url" target="_blank"
   rel="noopener">Open</a>
```

Page anchors

Give an element an id, then link to it with #. This jumps down the same page.

```
<a href="#end">Go down</a>
<h2 id="end">End</h2>
```

Email & phone links

mailto opens an email app. tel starts a phone call on mobile.

```
<a href="mailto:a@b.com">Email</a>
<a href="tel:+123456">Call</a>
```

Images

Add an image

img shows a picture. src is the file. alt is text for people who cannot see it. img has no closing tag.

```
<img src="cat.jpg" alt="A cat">
```

Size & loading

width and height keep the layout steady. loading="lazy" loads the image only when needed.

```
<img src="a.jpg" alt="" width="300"
     loading="lazy">
```

figure & figcaption

figure groups a picture with its caption.

```
<figure>
  <img src="a.jpg" alt="Sea">
  <figcaption>The sea</figcaption>
</figure>
```

picture & srcset

Give the browser different image files for different screens.

```
<picture>
  <source media="(max-width:600px)"
          srcset="small.jpg">
  <img src="big.jpg" alt="">
</picture>
```

Lists

Unordered list

ul makes a bullet list. Each item is an li.

```
<ul>
  <li>Apple</li>
  <li>Mango</li>
</ul>
```

Ordered list

ol makes a numbered list. Use start or type to change the numbers.

```
<ol start="1">
  <li>First</li>
  <li>Second</li>
</ol>
```

Description list

dl pairs a term (dt) with its meaning (dd). Good for glossaries.

```
<dl>
  <dt>HTML</dt>
  <dd>Page structure</dd>
</dl>
```

Nested lists

Put a list inside an li to make sub-items.

```
<ul>
  <li>Fruit
    <ul><li>Apple</li></ul>
  </li>
</ul>
```

Tables

Basic table

table holds rows (tr). Each row holds cells (td). th is a bold heading cell.

```
<table>
  <tr><th>Name</th><th>Age</th></tr>
  <tr><td>Ana</td><td>20</td></tr>
</table>
```

Table parts

thead, tbody and tfoot group the rows. caption gives the table a title.

```
<table>
  <caption>Scores</caption>
  <thead>…</thead>
  <tbody>…</tbody>
</table>
```

Merge cells

colspan joins columns. rowspan joins rows.

```
<td colspan="2">Wide cell</td>
<td rowspan="2">Tall cell</td>
```

Table tips

Use tables for data, never for page layout. Style borders with CSS.

```
table, td { border: 1px solid; }
```

Forms: Inputs

form tag

form collects user data. action is where to send it. method is get or post.

```
<form action="/send" method="post">
  …
</form>
```

input types

One input tag, many types: text, password, email, number, date, color, file, range and more.

```
<input type="text" name="user">
<input type="email">
<input type="date">
```

label

label names an input. Clicking the label selects the input. Match for with the input id.

```
<label for="n">Name</label>
<input id="n" type="text">
```

Buttons

button clicks do things. type="submit" sends the form. type="reset" clears it.

```
<button type="submit">Send</button>
```

Forms: More Controls

Choices

Radio buttons pick one option. Checkboxes pick many. Give radios the same name.

```
<input type="radio" name="g" value="m">
<input type="checkbox" name="news">
```

select & textarea

select makes a dropdown. textarea is a big box for long text.

```
<select name="c">
  <option value="in">India</option>
</select>
<textarea rows="3"></textarea>
```

Validation

The browser can check input for you with these attributes.

```
<input required minlength="3"
       maxlength="20" pattern="[a-z]+"
       placeholder="Name">
```

fieldset & datalist

fieldset groups related inputs with a legend. datalist suggests values while typing.

```
<fieldset>
  <legend>Contact</legend>
</fieldset>
```

Semantic HTML

Why semantic?

Semantic tags describe their meaning. This helps search engines, screen readers and other developers.

```
<header>…</header>
<main>…</main>
```

Page layout tags

header is the top. nav holds menus. main is the key content. footer is the bottom.

```
<header><nav>…</nav></header>
<main>…</main>
<footer>…</footer>
```

section, article, aside

section is a topic group. article stands alone, like a blog post. aside is side content.

```
<article>
  <h2>Post</h2>
</article>
<aside>Ad</aside>
```

Other semantic tags

time shows dates. address shows contact info. small is fine print. abbr explains short forms.

```
<time datetime="2026-10-05">Oct 5</time>
<abbr title="HyperText">HTML</abbr>
```

Block, Inline & Containers

Block elements

Block elements start on a new line and take the full width. Examples: div, p, h1, ul.

```
<div>Block</div>
```

Inline elements

Inline elements stay in the line and take only the space they need. Examples: span, a, strong, img.

```
Text <span>inline</span> text
```

div and span

div and span have no meaning. They are generic boxes for grouping and styling.

```
<div class="box">
  <span>Hi</span>
</div>
```

id and class

id is unique, used once per page. class can be used many times. CSS and JavaScript use them to find elements.

```
<p id="intro" class="big red">Hi</p>
```

Audio, Video & Embeds

Video

video plays movies. controls shows play and pause buttons. Add fallback text inside.

```
<video src="a.mp4" controls
       width="300"></video>
```

Audio

audio plays sound. It works like video.

```
<audio src="a.mp3" controls></audio>
```

iframe

iframe shows another web page inside yours, like a YouTube video or a map.

```
<iframe src="https://example.com"
        title="Demo"></iframe>
```

SVG & canvas

SVG draws sharp shapes that never blur. canvas draws with JavaScript, good for games and charts.

```
<svg width="50" height="50">
  <circle cx="25" cy="25" r="20"/>
</svg>
<canvas id="c"></canvas>
```

The Head & Meta Tags

title & charset

title shows on the browser tab. charset UTF-8 lets the page show every language and emoji.

```
<title>My Site</title>
<meta charset="UTF-8">
```

Viewport

This tag makes the page fit phone screens. Always include it.

```
<meta name="viewport"
  content="width=device-width, initial-scale=1">
```

SEO meta

Description is the text search engines show under your link.

```
<meta name="description"
  content="Learn HTML easily">
```

link & script

link adds CSS and a favicon. script adds JavaScript. Put script at the end of body or use defer.

```
<link rel="stylesheet" href="s.css">
<script src="app.js" defer></script>
```

Inline CSS

Style one element with the style attribute. Quick but hard to manage.

```
<p style="color:red">Red</p>
```

Internal CSS

Write CSS inside a style tag in head. Good for one page.

```
<style>
  p { color: blue; }
</style>
```

External CSS

Keep CSS in its own file and link to it. Best for real websites.

```
<link rel="stylesheet" href="style.css">
```

Adding JavaScript

JavaScript makes the page react to users. Add it with the script tag.

```
<script>
  alert("Hi");
</script>
```

Global Attributes

id, class, style, title

These work on every tag. title shows a tooltip when hovering.

```
<p id="a" class="b" title="Tip">Hi</p>
```

lang & dir

lang tells the language of the text. dir sets the reading direction.

```
<html lang="en">
<p dir="rtl">…</p>
```

data-* attributes

Store your own extra data on an element. JavaScript can read it.

```
<li data-id="7">Item</li>
```

hidden, tabindex, contenteditable

hidden hides an element. tabindex sets keyboard order. contenteditable lets people edit the text.

```
<p hidden>Secret</p>
<div contenteditable>Edit me</div>
```

Entities & Special Characters

What are entities?

Some characters have special meaning in HTML, like < and >. Write them as entities that start with & and end with ;.

```
&lt;  &gt;  &amp;  &quot;
```

Common entities

Use these for spaces and symbols.

```
   space that does not break
&copy;  &reg;  &euro;  &hearts;
```

Emoji & Unicode

With UTF-8 you can type emoji and any language directly.

```
<p>Hello 😀 नमस्ते</p>
```

Escaping code

To show a tag as text on the page, write its < and > as entities.

```
&lt;p&gt;Hi&lt;/p&gt;
```

Interactive Elements

details & summary

Makes a click-to-open box with no JavaScript.

```
<details>
  <summary>Click</summary>
  Hidden text
</details>
```

dialog

A popup window. Open it with JavaScript.

```
<dialog id="d">Hello</dialog>
<!-- d.showModal() -->
```

progress & meter

progress shows task progress. meter shows a value in a range, like disk space.

```
<progress value="40" max="100"></progress>
<meter value="6" max="10"></meter>
```

template & slot

template holds HTML that is not shown until JavaScript uses it. Used in web components.

```
<template>
  <p>Later</p>
</template>
```

Accessibility

Why it matters

Accessible pages work for people who use screen readers, keyboards or voice. It also helps SEO.

```
alt, labels, headings, contrast
```

alt text & labels

Describe every useful image. Connect every input to a label.

```
<img src="a.png" alt="Red apple">
```

Heading order

Use headings in order: h1, h2, h3. Do not skip levels just for size.

```
<h1>…</h1>
<h2>…</h2>
<h3>…</h3>
```

ARIA

ARIA attributes add meaning when plain HTML cannot. Use real tags first.

```
<button aria-label="Close">X</button>
<div role="alert">Error</div>
```

HTML5 APIs

Web storage

Save small data in the browser with localStorage.

```
localStorage.setItem("k", "v");
```

Geolocation

Ask the user for their location. They must allow it.

```
navigator.geolocation
  .getCurrentPosition(show);
```

Drag & drop

Make elements draggable with an attribute and events.

```
<div draggable="true">Drag me</div>
```

Web Workers & more

Workers run code in the background. Other APIs include history, notifications and fullscreen.

```
new Worker("w.js");
```

Best Practices & Key Takeaways

Write clean HTML

Close your tags. Use lowercase. Indent nested tags. Keep one idea per element.

```
<ul>
  <li>Neat</li>
</ul>
```

Use the right tag

Choose tags by meaning, not looks. Use CSS for looks.

```
<nav> not <div class="nav">
```

Check your work

Use the W3C validator and browser developer tools (F12) to find mistakes.

```
validator.w3.org
```

Learning path

1. Basic tags and text. 2. Links, images, lists, tables. 3. Forms. 4. Semantic layout. 5. Add CSS, then JavaScript. 6. Build small pages.

Save code in a file ending with .html and open it in your browser.